

The Parity Stack and the Identity Surface

Readability-Operator Families and Source-Context Modularity in Weak-Field Gravitational Scenes

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Abstract

A gravitational coherence surface, or GCS, is a declared contribution-context parity surface under a declared readability rule. Earlier work in this program emphasized a tidal-readable source-source construction using $Q_i \propto M_i/r_i^3$, which produced many-source parity networks in weak-field scenes. This paper shows that this historical branch is not the unique parity structure available to the field. It is the $k = 3$ member of a broader readability-operator family.

For scalar monopole supports of the form M/r^k , each falloff order defines a distinct parity layer. $k = 1$ gives potential-support parity, $k = 2$ gives acceleration-magnitude parity, $k = 3$ gives tidal-readable parity, and higher orders define derivative-sensitive layers. Together these layers form the parity stack. The stack is not assumed to be globally nested. It must be evaluated per declared scene, comparison mode, readability operator, and evaluation rule.

The paper also introduces mode discipline. Source-source parity foams answer which source locally outranks another source under a declared operator. Source-context GCS surfaces ask whether a declared contribution exceeds its derived context. These are not alternate views of the same object. They are different geometric questions. In a companion local solar neighborhood computational campaign, source-source topology changed substantially across $k = 1, 2, 3$, while the $k = 3$ source-context comparison left most of the declared grid context-dominated rather than source-owned.

The result does not modify general relativity and does not introduce new forces, membranes, horizons, or observables. It clarifies the declaration grammar required before parity structures can be computed, compared, or visualized. The GCS remains the identity-bearing closed source-context specialization within a broader operator-indexed parity grammar. Future extensions may compare scalar parity geometry with tidal eigenframe orientation, producing an eigenframe witness layer for scalar-tensor readability.

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1 Introduction

The gravitational coherence surface began with a question of identity.

Where does a declared gravitational contribution remain readable as itself against the gravitational context in which it is embedded?

The question is easy to misunderstand because gravity does not stop at a boundary. A mass does not cease to contribute to the field because a surface has been drawn around it. General relativity contains no material wall of the kind being discussed here, and this paper does not introduce one.

The boundary in question is a bookkeeping boundary.

Every gravitational model begins by declaring a scene. A source roster is chosen. Coordinates are adopted. A domain is selected. A parent or exterior context is named. A contribution becomes the object of analysis. A readability rule is chosen.

Only after those declarations does the GCS question become precise:

Under the chosen rule, where does this contribution remain more readable as itself than as part of its context?

The GCS answers that question. It is a declared contribution-context parity surface under a declared readability rule. It marks where the support assigned to a declared contribution equals the support assigned to its derived context [1].

This makes the GCS an identity and modularity surface. Inside a closed GCS, the declared contribution remains source-resolved under the chosen rule. At the surface, source and context have equal claim. Outside the surface, the contribution is increasingly assimilated into context.

The field remains continuous. The attribution changes.

Later work extended this logic from one contribution against one context to many-source parity networks [2]. In that branch, individual source contributions were compared against one another. The historical implementation used a tidal-readable scalar proportional to M_i/r_i^3 , evaluated for each source i at a field point x . Pairwise parity surfaces were computed where $Q_i = Q_j$. Retained portions of these surfaces formed many-source foams, seams, low-margin regions, and adjudication nodes.

That construction was productive. It gave the analysis a way to read weak-field scenes as relational objects rather than as unstructured field samples.

But operational success concealed a default.

The definition always required a declared readability rule. In practice, the working method began treating one readability rule as if it were the rule. The word “parity” came to mean the $k = 3$ tidal-readable source-source branch unless otherwise specified.

This was not a failure of the original definition. It was a methodological habit. A useful default became so familiar that the choice beneath it stopped being visible.

The present paper corrects that drift.

The central claim is:

Parity is not a single geometry. Parity is a declared comparison under a declared mode and readability operator.

Once this is made explicit, the historical $k = 3$ source-source foam is not discarded. It is located. It becomes one layer inside a broader readability-operator family.

This discovery does not demote the GCS. It clarifies why the GCS is special.

The source-source parity stack is a grammar of ranking. It tells us how source dominance changes when the declared operator changes. The GCS remains the identity-bearing source-context surface. It asks not merely which source wins against another source, but whether a declared contribution exceeds its derived context.

All GCS are parity structures. Not all parity structures are GCS.

That distinction is the load-bearing wall of this paper.

2 GCS Root and the Identity Surface

A gravitational coherence surface belongs to a declared scene.

It is not a universal shell possessed by a source in isolation. It is not an absolute boundary where a source ends. It is not a geometric halo that can be computed without asking what comparison is being made.

A GCS is generated by a contribution, a context, and a rule.

Let A be a declared gravitational contribution. Let K be its derived context. A readability rule assigns comparable supports to both:

$$Q_A(x) \tag{1}$$

and

$$Q_K(x). \tag{2}$$

The GCS condition is

$$Q_A(x) = Q_K(x). \tag{3}$$

Equivalently, define the residual

$$\Psi_A(x) = Q_A(x) - Q_K(x). \tag{4}$$

The GCS is the zero set

$$\Sigma_A = \{x : \Psi_A(x) = 0\}. \tag{5}$$

The meaning of this surface depends on the declaration. Change the contribution, the context, the source roster, the approximation regime, or the readability rule, and the surface may change. This dependence is not a weakness. It is the point.

A source does not carry one universal gravitational identity boundary. It has contribution-context boundaries under declared rules.

When such a boundary closes around a region, it can support a modular bookkeeping interpretation. The closed surface marks the place where source-resolved support and context support reach parity. It gives the model a disciplined way to say: inside this region, the contribution remains readable as itself under the declared rule; outside it, context has equal or greater claim.

This is why closure matters.

A closed source-context GCS can carry an identity interpretation. It encloses a domain in which the contribution remains modularly readable. An open surface, a clipped face, a seam, a node, or a low-margin patch may still be diagnostically important, but it does not automatically inherit the same identity meaning.

The parity-stack discovery expands the family of parity structures, but it does not flatten them into one category. Some parity structures are ranking maps. Some are transition supports. Some are junction witnesses. Some are gap diagnostics. A closed source-context GCS is special because it supports modular identity.

The distinction can be stated compactly:

Parity stack = grammar of readable comparison.
GCS = closed identity sentence inside that grammar.

This is the correct relationship between the new result and the original GCS concept. The parity stack does not replace the GCS. It gives the GCS a larger grammar in which its role becomes sharper.

3 Source-Source Ranking and Source-Context Identity

The most important distinction in the new grammar is mode.

A parity claim is incomplete unless it declares what kind of comparison is being made.

The two central modes are source-source and source-context.

In source-source mode, two source-readable supports are compared:

$$Q_{i,k}(x) = Q_{j,k}(x). \tag{6}$$

This asks which source outranks another source under the declared operator. In a many-source scene, the comparison can be performed across source pairs or against the local runner-up. The result is a dominance map. If every sampled point has a winner, the result is a tessellation: a source-source foam.

A source-source foam is useful. It tells us which source dominates the local ranking. It reveals adjacency, competition, retained faces, seams, and adjudication structures.

But a ranking is not the same thing as identity.

A source may be stronger than every other single source at a point and still contribute less than the rest of the scene combined. It may win every local duel while losing to the context as a whole.

Source-context mode asks the stronger question.

In source-context mode, a declared contribution A is compared against its derived context K :

$$Q_{A,k}(x) = Q_{K,k}(x). \quad (7)$$

If K is the sum or declared aggregation of all non- A contributions, then A must exceed that context to own the point under the chosen rule. This is not a tournament among individual sources. It is a test of whether the contribution remains modularly readable against the scene in which it is embedded.

Thus the two modes ask different questions:

Source-source mode asks: Which source is strongest among sources here?

Source-context mode asks: Does this declared contribution exceed its context here?

The difference is not semantic. It changes the geometry.

Source-source mode tends to produce complete ownership maps. Every point can have a winner. Source-context mode may produce sparse source-resolved regions inside a largely context-dominated field. This is not a defect. It is the point of the stricter question.

The GCS belongs to source-context mode. It is the surface where a contribution and its context reach parity. This is why source-context closure carries identity weight, while source-source dominance alone does not.

A source-source foam shows the competitive anatomy of a scene.

A source-context GCS shows where a contribution remains a coherent module.

Both are valuable.

They are different objects.

Table 1: Parity mode discipline.

Mode	Comparison	Question answered	Geometry produced	Identity strength
Source-source pair-wise	$Q_i = Q_j$	Which source outranks which source here?	Dominance foam, retained faces, adjacency network	Ranking only
Source-vs-derived context	$Q_A = Q_K$	Does this contribution exceed the rest of the scene?	GCS, source-resolved core	Strong if closed or closure-tested
Source-vs-declared context	$Q_A = Q_{context}$	Does this contribution exceed a declared support context?	Declared-context boundary	Scene-dependent
Group-vs-parent context	$Q_{group} = Q_{parent}$	Does the child or group exceed parent context?	Nesting handoff surface	Strong hand-off relevance if validated

4 The Scalar Parity Stack

Once mode is declared, the next question is operator.

The historical source-source foam used a tidal-readable scalar of the form

$$Q_i(x) \propto \frac{M_i}{r_i^3}. \quad (8)$$

This is now understood as the $k = 3$ member of a broader scalar family. In the weak-field point-source monopole approximation, define

$$Q_{i,k}(x) = \frac{GM_i}{r_i(x)^k}, \quad (9)$$

where M_i is the source mass, $r_i(x)$ is the distance from source i to the field point x , and k is the declared falloff order.

The source-source parity condition between sources i and j is

$$Q_{i,k}(x) = Q_{j,k}(x). \quad (10)$$

Substituting the scalar support gives

$$\frac{GM_i}{r_i^k} = \frac{GM_j}{r_j^k}. \quad (11)$$

The common factor G cancels:

$$\frac{M_i}{r_i^k} = \frac{M_j}{r_j^k}. \quad (12)$$

Rearranging gives the pairwise parity relation

$$\frac{r_i}{r_j} = \left(\frac{M_i}{M_j} \right)^{1/k}. \quad (13)$$

This equation defines the scalar source-source parity surface for the declared operator k .

The mass-scaling exponent is $1/k$. As k increases, mass dependence weakens. In the high- k limit, the surface approaches the geometric bisector between the two sources regardless of mass ratio.

Different k values therefore ask different physical questions.

At $k = 1$, the comparison is potential-support parity. This is the slowest-falloff branch. It compares potential support under a declared sign or magnitude convention. It should not be called a binding boundary unless an energy frame and zero convention have been separately declared.

At $k = 2$, the comparison is acceleration-magnitude parity. It asks where two sources contribute equal acceleration magnitude in the Newtonian weak-field sense. It is not a force-balance surface by default. Equal magnitudes do not imply zero net acceleration unless vector direction and frame conditions are also satisfied.

At $k = 3$, the comparison is tidal-readable parity. This is the historical branch. It has the strongest present interpretation because it connects to weak-field tidal structure, electric-Weyl interpretation, and geodesic deviation. In the current point-source monopole model, the tested source-attributed $k = 3$ tensor reductions are proportional to M/r^3 and therefore produce the same source-source parity surfaces.

At $k = 4$ and higher, the comparison becomes increasingly derivative-sensitive. These branches are lawful in principle, but they require their own computation and interpretation.

Together these layers form the scalar parity stack.

The stack is not assumed to be globally nested. In a clean two-source case, surfaces may be ordered in a simple way. In a many-source scene, retained faces may change source pair, topology may change, and the winner or runner-up may shift between k values. The stack is therefore evaluated per scene, per mode, per operator, and per evaluation rule.

The key point is not that every k -layer forms a neat shell inside every other layer. The key point is that changing k changes the declared readability question.

The historical $k = 3$ foam was not wrong. It answered one declared question.

The parity stack is the recognition that other lawful questions were always available.

Table 2: k -order interpretation.

k	Support	Disciplined name	Physical reading	Caution
1	M/r	Potential-support parity	Equal potential support	Convention-heavy. Not a binding boundary unless energy frame is declared.
2	M/r^2	Acceleration-magnitude parity	Equal acceleration magnitude	Not force balance unless vector and frame conditions are declared.
3	M/r^3	Tidal-readable parity	Tidal or electric-Weyl adjacent readability	Historical branch. Strongest current present interpretation.
4+	M/r^k	Higher derivative-sensitive parity	Higher derivative readability	Lawful in principle. Not yet canonical as computed LSN result.

5 Empirical Summary: Local Solar Neighborhood

Evidence status note. The following empirical summary refers to a companion local solar neighborhood computational campaign reported with this paper. At the time of this draft, the campaign results are treated as companion computational evidence, while final public verification depends on archiving the associated data products and reproducibility notes.

The parity-stack claim was evaluated in a local solar neighborhood scene containing 57 declared sources: 10 solar-system sources and 47 stellar sources. The goal of the campaign was not observational detection. It was to test whether the new declaration grammar changes the geometry computed from a weak-field scene.

It does.

The campaign compared source-source topology across $k = 1, 2, 3$; tested Sun source-context closure across k ; calibrated the pairwise gap formula in clean two-source corridors; examined $k = 1$ behavior at extreme mass ratios; checked source-attributed $k = 3$ tensor reduction equivalence; and compared source-source and source-context geometry at $k = 3$.

The reported results support the parity-stack architecture.

In source-source mode, topology changed substantially across $k = 1$, $k = 2$, and $k = 3$. The adjudication-node count rose strongly across the stack. The identity of the largest source domain changed between $k = 2$ and $k = 3$. Rare five-cell vertices present at $k = 3$ were absent at $k = 1$ and $k = 2$.

These are not cosmetic differences. They show that k -layers are not merely rescaled copies of one surface. Changing the readability operator changes the relational geometry returned by the field.

In source-context mode, closure was tested for the Sun along declared rays at $k = 1$, $k = 2$, and $k = 3$. Closure held in all three cases, but the geometry changed substantially. The $k = 1$ source-context boundary was smaller and more spherical than the $k = 3$ boundary.

This reverses the naive source-source intuition. In source-source mode, slow falloff can expand the dominance of massive individual sources. In source-context mode, slow falloff also strengthens the cumulative distant context. The Sun is not competing against a single runner-up. It is competing against the aggregate remainder of the scene. Under that stricter comparison, the low- k boundary can move inward.

The same operator family therefore behaves differently by mode:

In source-source mode, low k can expand massive-source dominance.

In source-context mode, low k can shrink source identity domains because aggregate context remains powerful.

Same scene. Same operator family. Opposite scaling behavior by mode.

The strongest result came from comparing source-source and source-context $k = 3$ geometry directly. Source-source mode assigned a dominant source throughout the sampled grid. Source-context mode did not. Most of the declared grid was context-dominated, meaning no single source exceeded the cumulative contribution of all others.

This is the empirical hinge of the paper.

A source-source foam can make a field look fully owned because every point has a winner. A source-context comparison can reveal that most of the same field is actually multi-source context, with only sparse regions where a single source exceeds the rest.

The same data can therefore support two different geometries because the questions are different.

A source-source foam is a map of local winners.

A source-context GCS is a test of modular identity.

The campaign also calibrated the pairwise gap formula in clean two-source corridors, bounded the $k = 1$ extreme-ratio pathology in the declared roster, and confirmed that the tested $k = 3$ source-attributed tensor reductions are equivalent in the current point-source monopole model.

Together, these results show that the parity stack is not a notation change. It is a measured operator family whose topology and interpretation depend on both k and mode.

Table 3: LSN source-source topology across k .

Metric	$k = 1$	$k = 2$	$k = 3$
Distinct dominant sources	27	30	32
Largest cell source	Sirius A	Sirius A	AlphaCen A
Parity vertices	164,091	244,336	261,037
Plateau borders	1,498	3,965	4,668
Adjudication nodes	5	59	105
5-cell vertices	0	0	2

Table 4: Source-source versus source-context $k = 3$ retention.

Source	Source-source voxels	Source-context voxels	Retention
AlphaCen A	279,247	2,185	0.78%
Sirius A	227,991	34,218	15.0%
Sun	198,316	29,533	14.9%
Lalande 21185	132,804	28,713	21.6%
Epsilon Eridani	74,943	25,177	33.6%
Lacaille 8760	55,694	24,277	43.6%

6 Gap Geometry

Once parity is operator-indexed, the space between layers becomes meaningful.

For a clean two-source monopole comparison, the position of a parity surface depends on mass ratio and k . Let two sources have mass ratio μ and separation d . Along the line connecting them, the normalized gap between parity surfaces at orders k_1 and k_2 is

$$\frac{\text{gap}(k_1, k_2)}{d} = \frac{1}{1 + \mu^{1/k_2}} - \frac{1}{1 + \mu^{1/k_1}}. \quad (14)$$

This formula follows directly from Eq. 13. It gives the distance between two declared operator layers in the same two-source branch.

The gap is not monotonic in mass ratio. It vanishes for equal masses, grows for moderate asymmetry, and shrinks again for extreme asymmetry as both surfaces move toward the smaller source. The gap is therefore a candidate diagnostic of local mass hierarchy, but not a simple invertible ruler.

The companion campaign confirmed the formula in clean two-source corridors. Where the pairwise branch was isolated from strong third-source contamination, numerical computation matched the analytical prediction to high precision.

Clean corridors were rare in the many-source local solar neighborhood scene. This is not a failure of the formula. It is another sign of the scene’s multi-source character. In most corridors, additional sources participate enough that the clean two-body branch is no longer the right object to measure.

Gap geometry therefore has two legitimate but distinct uses.

As pairwise algebra, it is exact under the declared monopole model.

As retained many-source geometry, it is conditional. The relevant source pair must remain stable, third-source contamination must be controlled, and the surfaces being compared must correspond to one another.

This distinction preserves the diagnostic without exaggerating it. Gap geometry is useful, but only when its branch is declared.

7 Continuous k

The parity stack was first sampled at integer values: $k = 1$, $k = 2$, $k = 3$, and so on. But the scalar monopole equation does not require k to be an integer.

For real $k > 0$, the source-source parity condition

$$\frac{M_i}{r_i^k} = \frac{M_j}{r_j^k} \quad (15)$$

remains well-defined away from singular and degenerate cases. Taking logarithms gives

$$\ln \left(\frac{M_i}{M_j} \right) = k \ln \left(\frac{r_i}{r_j} \right). \quad (16)$$

For a fixed source pair and a nondegenerate point, one can solve for an effective readability order:

$$k_{\text{eff}}(x; i, j) = \frac{\ln(M_i/M_j)}{\ln(r_i/r_j)}. \quad (17)$$

Within a fixed source-pair branch, the region between integer k surfaces is therefore not merely a visual interpolation. It is filled by a continuous family of lawful parity surfaces.

The integer stack is a sampled skeleton of a continuous operator family. This turns k from a small list of named diagnostics into a continuous readability parameter.

This statement requires discipline. Continuous k is not a new physical dimension. It is not a claim that nature contains a hidden k -axis. It is a diagnostic parameter inside a declared family of readability operators.

It is also not automatically global. In a many-source retained foam, the relevant source pair may change across the scene. The local runner-up may change as k changes. Retained faces may appear, disappear, split, or merge. Effective k is therefore branch-indexed.

Within a declared branch, however, continuous k gives the framework a powerful new question:

At what k does this structure become visible?

A retained feature that appears at $k = 3$ but not at $k = 2$ may have a critical value k_c between them. A junction that becomes stable at lower k may be more robust than one that appears only at higher derivative-sensitive orders. A rare node can be classified not only by how many regions meet there, but by the readability order at which the relationship first becomes visible.

This turns the stack from a set of snapshots into a feature space.

A parity object can now carry more than a location and source roster. It can carry a visibility threshold in operator space.

8 Normalized Margin Geometry

Exact parity is the zero surface. But a scene also contains near-parity structure: regions where two supports are not equal but are close enough to matter.

A naive way to describe this would be to offset the equality condition by an absolute amount:

$$Q_i = Q_j + \tau. \quad (18)$$

This is poorly suited to multiscale gravitational scenes. The quantity τ has field units. The same absolute offset can mean very different things near a strong source and far away from it. It does not provide a scale-stable measure of competition.

A better margin is relational.

Define the log-ratio residual

$$\rho_{ij,k}(x) = \ln \left(\frac{Q_{i,k}(x)}{Q_{j,k}(x)} \right). \quad (19)$$

Exact parity occurs when

$$\rho_{ij,k} = 0. \quad (20)$$

A fractional offset

$$\frac{Q_{i,k}}{Q_{j,k}} = 1 + \epsilon \quad (21)$$

corresponds to

$$\rho_{ij,k} = \ln(1 + \epsilon). \quad (22)$$

For small ϵ , ρ is approximately ϵ .

This log-ratio margin supplies a normalized way to discuss near-parity shells, low-margin regions, confidence bands, and transition thickness. It says not only where two supports are equal, but how strongly one outranks the other.

For a fixed source pair, k and ρ define a two-parameter family of level sets:

$$\ln \left(\frac{M_i}{M_j} \right) - k \ln \left(\frac{r_i}{r_j} \right) - \rho_0 = 0. \quad (23)$$

This does not automatically create a global coordinate system for the scene. It becomes a local chart only under declared conditions: fixed source pair, fixed mode, stable branch identity, nonzero residual gradient, and no topology change inside the chart.

That limitation is not a defect. It is the correct scale of the claim.

The framework does not need false global coordinates. It needs disciplined local charts that say exactly what is being compared.

Normalized margin geometry also connects directly to source-context analysis. A source-source foam can assign a winner even when the winner's margin is small. A source-context comparison can reveal that the apparent winner is embedded in a larger context. Together, margin and mode give a more honest description of field readability than either alone.

9 Adjudication Nodes as Relational Events

Many-source parity networks do not contain only surfaces.

They also contain seams, retained borders, low-margin regions, and points where several domains meet. This paper calls some of these higher-order locations adjudication nodes: places where multiple sources participate in simultaneous or near-simultaneous parity structure.

In an integer- k computation, an adjudication node can appear as a point in space. It is tempting to treat it as a spatial fixture. But the parity-stack view suggests a deeper interpretation.

An adjudication node is not merely a place. It is a relationship with a current address.

The node exists because a particular set of sources, under a particular mode and operator, supports a local multi-way transition. Change the source configuration and the node may migrate. Change the operator and the node may appear or vanish. Change the mode and the same spatial region may no longer qualify as an adjudication structure.

The position is where the relationship manifests. The relationship is the object.

This reframing matters because it changes how such nodes should be recorded. A node is not fully described by x , y , and z . It needs a relational record.

A disciplined node record should include source set, parity mode, readability operator, observed k , node type, retained-face relations, margin structure, critical k if known, stability under perturbation, and claim status.

The critical- k extension makes this especially useful. A node observed at $k = 3$ but absent at $k = 2$ may have a birth threshold between them. Another node may persist across a wide interval of k . These are different kinds of relational events. One is fragile, visible only under a higher derivative-sensitive operator. The other is robust across multiple readability orders.

This gives the framework a possible taxonomy of junctions.

A node may be classified not only by how many regions meet there, but by the operator interval over which it survives. A five-cell node that appears only near $k = 3$ is not the same kind of event as a four-cell node stable from $k = 1.8$ to $k = 3.5$. The stack gives node topology an operator coordinate.

In future time-dependent scenes, such a node may trace a worldline as the source configuration evolves. It may be born, drift, split, merge, or die. Even in a static scene, the conceptual correction is useful:

An adjudication node is a relational event sampled on a slice.

This does not make the node a physical particle, causal event, or observable. It makes the bookkeeping more honest. The analysis is not only recording where a junction appears. It is recording what relationship had to be true for that junction to appear there.

10 Eigenframe Witness Geometry

The scalar parity stack ranks contribution magnitude under declared operators. It tells us who wins, or whether a contribution exceeds context, according to a chosen scalar support.

But weak-field tidal structure also has direction.

The tidal tensor carries eigenvalues and eigenvectors. The eigenvalues describe stretching and compression magnitudes. The eigenvectors define principal axes of tidal deformation [7, 8, 9]. This suggests a natural extension of the parity-stack program: compare scalar parity geometry with tidal eigenframe orientation.

This extension must be handled carefully.

In the current point-source monopole source model, source-attributed eigenvalue parity does not create an independent new foam. For a point-source monopole tidal tensor, the tested scalar reductions are proportional to M/r^3 . Frobenius norm, maximum eigenvalue magnitude, and source-radial projection differ from the historical Q scaffold by source-independent constants. When equality is tested between sources, those constants cancel. Thus source-attributed eigenvalue parity collapses to the already established $k = 3$ scalar branch.

The new tensorial information is not eigenvalue equality.

It is orientation.

Let a retained k -parity face be defined by the residual

$$\Psi_{ij,k} = Q_{i,k} - Q_{j,k}. \quad (24)$$

Where the surface is smooth, define its normal

$$n_{ij,k} = \frac{\nabla \Psi_{ij,k}}{\|\nabla \Psi_{ij,k}\|}. \quad (25)$$

Let $e_{0,\text{total}}(x)$ be the dominant eigenaxis of the total weak-field tidal tensor at the same location. Because eigenvectors are axes rather than oriented arrows, e and $-e$ should be treated as equivalent.

Define the eigenframe divergence angle

$$\delta_{ij,k}(x) = \arccos(|e_{0,\text{total}}(x) \cdot n_{ij,k}(x)|). \quad (26)$$

This angle asks whether the scalar ranking geometry agrees with the directional geometry of the total tidal field.

If δ is near zero, the total tidal eigenaxis aligns with the normal of the scalar parity surface. If δ is near $\pi/2$, the eigenaxis lies tangent to that surface or is otherwise decoupled from the scalar ranking direction.

At clean two-source boundaries, some degree of alignment may be expected. At adjudication nodes, low-margin regions, and context-dominated volumes, the answer is open. The tensor field may become ambiguous where scalar attribution becomes ambiguous. Or it may retain a coherent direction even where source ranking is contested.

The second case would be the more interesting result. It would mean the field's directional organization carries information not captured by source dominance alone.

This is why the proposed object is called an eigenframe witness layer, not an eigenvector parity foam. Its job is not to replace the scalar stack. Its job is to witness how the scalar stack sits inside the directional structure of the field.

The scalar parity stack says who wins under a declared operator.

The eigenframe witness says which way the field is trying to deform.

Where those agree, the field repeats itself in two languages. Where they diverge, the comparison reveals a genuine diagnostic layer.

This paper does not claim that the eigenframe witness has been computed or validated. It is included as the next logical extension of the theory.

11 Visualization as Conceptual Stress Test

The parity-stack clarification did not arise from algebra alone.

It emerged because a visual modeling problem refused to go away.

A weak-field diagram can easily become misleading if it does not declare what geometric object it represents. A surface drawn as “the boundary” may suggest a physical wall, a sphere of influence, or a force cutoff unless the comparison behind that surface is made explicit. In this case, the visual question was simple:

What should the visible structure terminate against?

That question exposed the hidden default. If a diagram terminates at parity, then one must ask:

Parity of what? Against whom? Under which operator?

The working method did not yet have a sufficiently explicit declaration. It had a habit. Parity meant the $k = 3$ source-source Q branch unless the question was reopened.

The visualization problem forced that reopening.

The visual model did not decide the result. The definitions, calculations, and comparison tests did. But the attempt to draw the structure exposed what the calculation had to specify.

This is a methodological lesson for public-facing gravitational scene analysis. Visualization is not evidence by itself. A clear diagram does not certify a physical object. But a disciplined visual model can reveal missing declarations. It can ask whether the object shown is source-source or source-context, which operator is being used, which context is being compared, and what claim is being made.

In this case, visual modeling exposed an operator ambiguity at the heart of the analysis. The attempt to draw a lawful weak-field structure forced the question of what kind of parity the drawing was supposed to represent. That question produced the parity stack.

The rule is:

Visualization does not decide. But visualization can ask the question that reveals what the calculation must specify.

12 Claim Ladder

The claims of this paper should be read in levels.

The first three levels are definitional: parity requires declared mode and operator; source-source k -foam and source-context k -GCS are distinct object classes; scalar monopole supports M/r^k form an operator-indexed integer stack within the declared model.

The fourth level is empirical: in the companion LSN campaign, k -layer topology and mode behavior differ substantially.

The fifth and sixth levels are mathematical and diagnostic extensions: continuous real $k > 0$ and normalized log-ratio margin.

The seventh and eighth levels are future programs: critical- k node taxonomy and eigenframe witness geometry.

Horizon topics such as intersection worldlines, node persistence, galaxy-scale topology signals, and frame-dragging diagnostics require separate theory and computation. They are not claims of this paper.

Table 5: Claim ladder.

Level	Claim	Status
1	Parity requires declared mode and operator.	Core claim
2	Source-source k -foam and source-context k -GCS are distinct object classes.	Core claim
3	Scalar monopole supports M/r^k form an integer parity stack.	Canonical within declared model
4	LSN k -layer topology and mode behavior differ substantially.	Empirical result, pending raw-data verification if data products remain external
5	Scalar monopole family extends to continuous real $k > 0$.	Mathematical extension
6	Normalized log-ratio margin describes near-parity structure.	Mathematical and diagnostic extension
7	Critical k values classify node and topology transitions.	Candidate program
8	Eigenframe witness geometry compares scalar ranking and tensor orientation.	Future computation target

13 Non-Claims

This paper does not modify general relativity. It does not introduce a new force, field equation, particle, membrane, horizon, or causal boundary.

It does not claim that parity surfaces are physical walls. It does not claim that source-source foams are GCS surfaces. It does not demote the GCS.

It does not claim that the parity stack is globally nested, that k is a physical dimension, or that all eigenvector constructions produce closed surfaces.

It does not claim observational detection, a galaxy-rotation explanation, a frame-dragging diagnostic, or that visualization is evidence.

The claim is narrower and stronger:

In declared weak-field scenes, parity is a mode- and operator-indexed diagnostic grammar.
The GCS is the identity-bearing source-context specialization within that grammar.

14 Conclusion

The historical parity foam was not wrong. It was underspecified.

It was the $k = 3$ tidal-readable source-source branch of a larger readability-operator family. Once this is recognized, the field reveals a richer grammar: potential-support parity, acceleration-magnitude parity, tidal-readable parity, higher-order layers, near-parity margins, topology changes across k , and mode-dependent distinctions between ranking and identity.

The GCS remains central. It is not replaced by the parity stack. It is clarified by it.

A source-source foam tells us who wins locally under a declared operator. A source-context GCS tells us whether a declared contribution remains stronger than its derived context. That is the difference between ranking and identity.

The parity stack gives the framework a broader language for comparison. The GCS gives it a root language for modularity.

The continuous- k extension turns integer layers into a sampled operator family. Normalized margin geometry turns exact parity into a neighborhood of near-parity structure. Adjudication nodes become relational events rather than mere spatial fixtures. The eigenframe witness points toward a tensorial extension in which scalar ranking can be compared with directional tidal organization.

Together these ideas define a more mature analysis grammar:

Declare the scene. Declare the mode. Declare the operator. Declare the context.
Compute. Validate. Visualize only after the calculation is specified.

This is the step from finding surfaces to reading scenes.

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